

Reposo T2

Métodos Matemáticos Para La Física

$$1. y'' - 4y' + 4 = 0$$

Ec. Característica: $r^2 - 4r + 4 = 0 \rightarrow (r-2)^2 = 0 \rightarrow r = 2$ con $h = 2$

$$y(x) = (c_1 + c_2 t) e^{2t}$$

$$2. 2y'' + 4y' + 7y = x^2$$

Ecuación Característica: $2r^2 + 4r + 7 = 0 \rightarrow r^2 + 2r + \frac{7}{2} = 0$, $(r+1)^2 - 1 + \frac{7}{2} = 0$, $(r+1)^2 = -\frac{5}{2}$, $r = -1 \pm \frac{\sqrt{10}}{2} i$

$$y_c(x) = e^{-x} \left(\cos\left(\frac{\sqrt{10}}{2} x\right) + \sin\left(\frac{\sqrt{10}}{2} x\right) \right)$$

$$y_p = A + Bx + Cx^2, y' = 2Cx + B, y'' = 2C$$

$$2(2C) + 4(2Cx + B) + 7(Cx^2 + Bx + A) = x^2 \rightarrow 4C + 8Cx + 4B + 7Cx^2 + 7Bx + 7A$$

$$7A + 4B + 4C = 0 \rightarrow 7A - \frac{32}{49} + \frac{1}{4} = 0 \rightarrow 7A = \frac{4}{49} \rightarrow A = \frac{4}{343}$$

$$7B + 8C = 0 \rightarrow 7B = -\frac{8}{7} \rightarrow B = -\frac{8}{49}$$

$$7C = 1 \rightarrow C = \frac{1}{7}$$

$$y_p = \frac{1}{343} (4 - 56x + 49x^2), y(x) = y_c + y_p = e^{-x} \left(c_1 \cos\left(\frac{\sqrt{10}}{2} x\right) + c_2 \sin\left(\frac{\sqrt{10}}{2} x\right) \right) + \frac{1}{343} (4 - 56x + 49x^2)$$

$$3. y''' - y = e^x + 7$$

Ecuación Característica: $r^3 - 1 = 0 \rightarrow r = 1$

$$r-1 \overline{\begin{array}{r} r^2+r+1 \\ r^3-r^2 \\ \hline r^2-r \end{array}}$$

$$(r^3 - 1) = (r-1)(r^2 + r + 1)$$

$$(r + 1/2)^2 + 3/4 = 0 \rightarrow r = -1/2 \pm \frac{\sqrt{3}}{2} i$$

$$y_c = c_1 e^x + e^{-x/2} \left(c_2 \cos\left(\frac{\sqrt{3}}{2} x\right) + c_3 \sin\left(\frac{\sqrt{3}}{2} x\right) \right)$$

$$\begin{array}{r} r-1 \\ r-\frac{1}{2} \\ \hline 0 \end{array}$$

$$y_p(x) = Ax e^x + B \quad y'_p = Ae^x + A x e^x, y''_p = 2Ae^x + A x e^x, y'''_p = 3Ae^x + A x e^x$$

$$Ax e^x + 3Ae^x - A x e^x - B = e^x + 7 \rightarrow 3Ae^x - B = e^x + 7, 3A = 1 \rightarrow A = \frac{1}{3}, B = -7$$

$$y_p = \frac{1}{3} x e^x - 7 \rightarrow y(x) = C_1 e^x + C_2 \left(C_2 \cos\left(\frac{\sqrt{10}}{2} x\right) + C_3 \sin\left(\frac{\sqrt{10}}{2} x\right) \right) + \frac{1}{3} x e^x - 7$$

$$4. y'' + y = \sin(x) + x \cos(x)$$

$$\text{Ec. Característica: } r^2 + 1 = 0 \rightarrow r = \pm i \rightarrow y_c(x) = \cos(x) + \sin(x)$$

Encontramos solución particular:

$$\sin(x) \rightarrow A \sin(x) + B \cos(x) \rightarrow \text{Hay duplicación} \rightarrow A x \sin(x) + B x \cos(x)$$

$$x \cos(x) \rightarrow C x \cos(x) + D x \sin(x) + E \cos(x) + F \sin(x) \rightarrow \text{Hay Duplicación} \rightarrow \text{Multiplicar por } x$$

$$y_p = \underbrace{(A' + E)}_{A'} x \sin(x) + \underbrace{(B' + F)}_{B'} x \cos(x) + C x^2 \sin(x) + D x^2 \cos(x)$$

$$y_p = x(A' \sin(x) + B' \cos(x)) + x^2(C \sin(x) + D \cos(x))$$

$$y'_p = (A' \sin(x) + B' \cos(x)) + x(A' \cos(x) - B' \sin(x)) + 2x(C \sin(x) + D \cos(x)) + x^2(C \cos(x) - D \sin(x))$$

$$y''_p = 2(A' \cos(x) - B' \sin(x)) - x(A' \sin(x) + B' \cos(x)) + 2(C \cos(x) + D \sin(x)) + 4x(C \cos(x) - D \sin(x)) - x^2(C \sin(x) + D \cos(x))$$

$$y''_p = 2((A' + D) \cos(x) + (C - B') \sin(x)) + 4x((C - \frac{B'}{2}) \cos(x) - (\frac{A'}{2} + D) \sin(x)) - x^2(C \sin(x) + D \cos(x))$$

$$2((A' + D) \cos(x) + (C - B') \sin(x)) + 4x((C - \frac{B'}{2}) \cos(x) - (\frac{A'}{2} + D) \sin(x)) - x^2(C \sin(x) + D \cos(x)) + x(A' \sin(x) + B' \cos(x)) + x^2(C \sin(x) + D \cos(x)) = \sin(x) + x \cos(x)$$

$$2(C - B') = 1 \rightarrow 2C - 2B' = 1 \rightarrow C = \frac{1}{2} + B' \rightarrow C = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$4(C - \frac{B'}{2}) + B' = 1 \rightarrow 2 + 4B' - B' + B' = 1 \rightarrow 4B' = -3 \rightarrow B' = -\frac{3}{4}$$

$$2(A' + D) = 0 \rightarrow A' = -D \rightarrow A = 0$$

$$A' + 4D - A = 0 \rightarrow 2D = 0 \rightarrow D = 0$$

$$y_p(x) = -\frac{1}{4}x \cos(x) + \frac{1}{4}x^2 \sin(x)$$

$$y(x) = a_1 \cos(x) + a_2 \sin(x) + \frac{x^2}{4} \sin(x) - \frac{x}{4} \cos(x)$$

$$5. \quad y'''' - 5y'' + 4y = e^x - x e^{2x}$$

$$\text{Ecuación Característica: } r^4 - 5r^2 + 4 = 0 \rightarrow n = r^2 \rightarrow n^2 - 5n + 4 = 0 \rightarrow (n-4)(n-1) = 0$$

$n = 4, 1$

$$n = r^2 \rightarrow r = \sqrt{n} \quad r = \pm 2, \pm 1 \rightarrow y_c = C_1 e^{2x} + C_2 e^{-2x} + C_3 e^x + C_4 e^{-x}$$

Busco y_p :

$$e^x: \text{ Hay Duplicación } \rightarrow A x e^x$$

$$x e^{2x}: \text{ No hay Duplicación } \rightarrow B x e^{2x} + C e^{2x} \rightarrow \text{ Ahora si } \rightarrow B x^2 e^{2x} + C x e^{2x}$$

$$y_p = A x e^x + B x^2 e^{2x} + C x e^x$$

$$y''_p(x) = A(e^x + 2e^x) + B(4e^{2x}x^2 + 8e^{2x}x + 2e^{2x}) + C(4e^{2x}x + 4e^{2x})$$

$$y''''_p(x) = A(e^x + 4e^x) + B(16e^{2x}x^2 + 64e^{2x}x + 48e^{2x}) + C(16e^{2x} + 32e^{2x})$$

$$A(e^x + 4e^x) + B(16e^{2x}x^2 + 64e^{2x}x + 48e^{2x}) + C(16e^{2x} + 32e^{2x})$$

$$-5(A(e^x + 2e^x) + B(4e^{2x}x^2 + 8e^{2x}x + 2e^{2x}) + C(4e^{2x}x + 4e^{2x}))$$

$$4(A(e^x + 2e^x) + B(4e^{2x}x^2 + 8e^{2x}x + 2e^{2x}) + C(4e^{2x}x + 4e^{2x})) = e^x - x e^{2x}$$

$$A = -1/6, \quad B = -1/24, \quad C = \frac{19}{144}$$

$$y(x) = C_1 e^{2x} + C_2 e^{-2x} + C_3 e^x + C_4 e^{-x} - \frac{e^x x}{6} - \frac{e^{2x} x^2}{24} + \frac{19 e^{2x} x}{144}$$

